



NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY GREATER NOIDA-201306

(An Autonomous Institute)

School of Computer Science & Information Technology

Subject Name: Operating Systems		L-T-P [2-0-0]
Subject Code: BCSE0303		Applicable in Department: CSE/CSE (R)/ IT/ CSE(Twin)/IT(Twin)/CSE(Prof)/IT(Prof)/ CS/ DS /IOT/AI/ML/AI(AI/ML(TWIN)/AI(TWIN)/CYS/ M&C/ M.Tech (Int.)
Pre-requisites of the Subject: Basic understanding of C/ C++ Programming, Data structures & algorithms, Computer Organization & architectures.		
Course Objective- The objective of the course is to provide a foundational understanding of operating system concepts, including system architecture, process and thread management, concurrency, deadlock, resource management, memory and file systems, Linux shell scripting, and an introduction to virtualization and distributed systems.		
Course Outcomes (CO)		
Course outcome: After completion of this course, students will be able to:		Bloom's Knowledge Level(KL)
CO 1	Understand operating systems architecture and types, and use the Linux CLI for basic Operations.	K2
CO2	Implement the CPU scheduling algorithms along with uses of multithreading models.	K4

CO3	Implement concurrency control, process synchronization techniques, and deadlock handling methods.					K4
CO4	Implement memory management strategies and page replacement algorithms to optimize system performance.					K4
CO5	Analyze file systems and configure distributed systems and virtual machines in modern operating systems.					K4
Syllabus						
Unit No	Module Name	Topic covered	Pedagogy	Lecture Required (L+P)	Practical/ Assignment/ Lab Nos	CO Mapping
1 Fundamentals & Shell scripting	Module 1.1: Fundamentals of Operating Systems	Fundamentals of Operating Systems, Overview of Operating Systems, Generations of OS, Operating system architecture, Interrupt handling, System call and kernel, Types of Operating System: Batch OS, Multiprogramming OS, Multitasking OS, Multiprocessor OS, Real time OS.	Class room Teaching, Smart Board, PPT.	2	Assignment 1	CO1
	Module 1.2: Shell Scripting in Linux	Shell Scripting in Linux, Introduction to Linux Operating System & Architecture, Basic Command Line Interface (CLI) Operations in Linux, Shell Scripting Basics: Variables, Control Structures, Functions Case Study: Automating system administration tasks using shell scripts in Ubuntu/Linux (e.g., backup scheduling).	Class room Teaching, Smart Board, PPT.	2		CO1

2 Process & Thread Manage ment	Module 2.1: Process Management and CPU Scheduling	Process Management: - Process, Transition Diagram, Process Control Block (PCB), Types of Schedulers: Long Term, Mid Term, Short Term Scheduler. CPU Scheduling- Pre-emptive and Non-Pre-emptive Algorithm (FCFS, SJF, SRTF, Non-Pre-emptive Priority, Pre-emptive Priority, Round Robin, Multilevel Queue Scheduling and Multilevel Feedback Queue Scheduling) .	Class room Teaching, Smart Board, PPT.	4	Practical 4 to 9	CO2
	Module 2.2: Thread	Thread: - Processes vs Threads, Thread states, Benefits of threads, Types of threads, Multithread Model, Concept of Hyper-Threading Case Study: - Analyse and implement CPU Scheduling in Real-Time Embedded Systems and RTOS	Class room Teaching, Smart Board, PPT.	4	Assignment 1	CO2
3 Concur rency and Deadlo ck Manag ement	Module 3.1: Concurrency, Process Synchronizati on	Concurrency: Introduction of Concurrency, Types of Process, Race Condition, Critical Section, Inter Process Communication, Producer consumer problem. Process Synchronization: Lock variable, Peterson's Solution, Strict alternation, Lamport Bakery Solution, Test and set lock, Semaphore-counting, binary and monitor, Classical Problem of Synchronization: - Bound Buffer, Dining Philosopher, Reader writer, Sleeping barber.	Class room Teaching, Smart Board, PPT.	4	Assignment 2, Practical 10,11,12	CO3

	Module 3.2: Deadlock	Deadlock: Deadlock, Deadlock characterization, Deadlock Prevention, Deadlock Avoidance: Bankers Algorithms, Deadlock Detection, Recovery from Deadlock. Case Study: Deadlock avoidance in database transaction management systems like Oracle or MySQL.	Class room Teaching, Smart Board, PPT.	4	Practical 13,14	CO3
4 Memory Management	Module 4.1: Memory Management	Memory Management: - Memory Management function, Loading and linking Address Binding, Memory management techniques, Contiguous technique- Fixed Partitions, variable partitions, Memory Allocation: Allocation Strategies (First Fit, Best Fit, and Worst Fit), Non-contiguous, Paging, Segmentation, Segmented paging	Class room Teaching, Smart Board, PPT.	4	Assignment 3	CO4
	Module 4.2: Virtual memory	Virtual Memory:- Virtual Memory Concepts, Demand Paging, Performance of Demand Paging, Page Replacement Algorithms: FIFO, LRU, Optimal and LFU, Belady's Anomaly, Thrashing Case Study: Virtual memory management in modern OS like Windows 10 and how paging impacts performance.	Class room Teaching, Smart Board, PPT.	4	Practical 21 to 23	CO4
5 File Management & Modern Operating System	Module 5.1:	File Management: - Access Mechanism, File Allocation Method, Free Space Management: -Bit Vector, Linked List. DISK: Disk Architecture, HDD vs SSD, Disk Scheduling Algorithms	Class room Teaching, Smart Board, PPT.	2	Practical 22 to 24	CO5
	Module 5.2:	Modern Operating System: -Overview of modern operating system, Modern OS features: Multitasking, virtualization, security, scalability, Shared Memory concepts, Distributed system, Parallel system & its architecture, Virtual machines – hypervisor, Introduction to GPU Case Study: Large File Storage in a Distributed Manner.	Class room Teaching, Smart Board, PPT.	2	Assignment 3	CO5
Total				32		

Textbooks	
Sr No	Book Details
1.	Abraham Silberschatz, Peter Baer Galvin and Greg Gagne” Operating System Concepts Essentials” , Willey Publication,10th Edition,2018.
2.	Marks G. Sobell “A practical guide to Linux: Commands, Editors and Shell Programming”, CreateSpace Independent Publishing Platform, 4 th Edition,2017.
3.	Jason Cannon “LINUX for beginners”, 1stEdition,2014
Reference Books	
Sr No	Book Details
1.	William Stallings “Operating Systems: Internals and Design Principles”, Pearson Education , 9 th Edition, 2019.
2.	Charles Patrick Crowley, “Operating System: A Design-oriented Approach” , McGraw Hill Education ,2017.
3.	Ganesh Naik “Learning Linux Shell Scripting”, Packt Publishing ,2 nd Edition 2018.
Links	
Unit 1	CS162 Lecture 1: What is an Operating System? (youtube.com) Operating System #01 Introduction to OS, its Roles & Types (youtube.com) Operating System #14 What is an Interrupt? Types of Interrupts - YouTube https://www.youtube.com/watch?v=akU1Ji8Vzdk&list=PLbMVogVj5nJRa3VKt_eyZdJ_DitCz1cvQ https://www.youtube.com/watch?v=rRGCGZ6OHw8&list=PLbMVogVj5nJRa3VKt_eyZdJ_DitCz1cvQ&index=2
Unit 2	Operating System #03 Programs & Processes, System Calls, OS Structure (youtube.com) Operating System #18 CPU Scheduling: FCFS, SJF, SRTF, Round Robin - YouTube Operating System #19 Priority Scheduling Algorithms, Multilevel Queues - YouTube Operating System #20 Multi Processor Scheduling (youtube.com)

	Operating System #33 Threads: Thread Model, Thread vs Process, pthread library (youtube.com) Operating System #34 Threads: User level & Kernel level thread, Threading issues (youtube.com) https://www.youtube.com/watch?v=3eG27YUbzyM&list=PLbMVogVj5nJRa3VKt_eyZdJ_DitCz1cvQ&index=3
Unit 3	CS162: Lecture 6: Synchronization 1: Concurrency and Mutual Exclusion (youtube.com) CS162: Lecture 6.5: Concurrency and Mutual Exclusion (Supplemental) (youtube.com) Operating System #04 CPU Sharing, Race Conditions, Synchronization, CPU Scheduling (youtube.com) Operating System #26 Bakery Algorithm - YouTube Operating System #27 Hardware Locks: Spinlock & its Usage (youtube.com) Operating System #31 Deadlocks: Deadlock Detection & Recovery (youtube.com)
Unit 4	Operating System #05 Memory Management: Process, Fragmentation, Deallocation, (youtube.com) Operating System #06 Virtual Memory & Demand Paging in Operating Systems (youtube.com) Operating System #07 MMU Mapping How Virtual Memory Works? - YouTube
Unit 5	https://www.youtube.com/watch?v=qbQCQ0U6H0o https://www.youtube.com/watch?v=SnKgEuUfV4k https://www.youtube.com/watch?v=cVFyK1f5lDw https://www.youtube.com/watch?v=Z0Vkm9faoM&list=PLbMVogVj5nJRa3VKt_eyZdJ_DitCz1cvQ&index=4 https://www.youtube.com/watch?v=_BtDcroOTSA CUDA Programming Course – High-Performance Computing with GPUs



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Subject Name: Operating System Lab		L-T-P [0-0-4]
Subject Code: BCSE0353		Applicable in Department: CSE/CSE-R/IT/CS/AI/AIML/ IOT/DS/CYS
Pre-requisite of Subject: Basic knowledge of computer fundamentals, C programming, Data structure and Computer organization.		
Lab Experiments		
Course Objective: The course aims to provide hands-on experience with Linux, shell programming and analyzing core OS concepts such as process scheduling, synchronization, memory management, disk scheduling and familiarization with modern OS tools.		
Course Outcomes (CO)		
Course outcome: After completion of this course students will be able to:		Bloom’s Knowledge Level(KL)
CO1	Execute basic Linux commands and shell scripts to automate file management and system administration tasks.	K3
CO2	Implement and compare various CPU scheduling algorithms, process synchronization solutions using semaphores and deadlock handling algorithms.	K4
CO3	Simulate memory allocation techniques and page replacement algorithms, disk management strategies and explore modern OS features including virtualization and distributed computing.	K4

List of Practical		
Sr. No	Program Title	CO Mapping
1.	Implementation of Linux Commands Introduction of Unix/Linux Operating system and their architecture Display system information using uname, hostname, and date etc. File operations using cat, touch, cp, mv, rm, and chmod ,umask etc. Create, view, and navigate directories using mkdir, rmdir, cd, pwd, ls etc. Disk Commands df,du,mount,unmount,mkfs,fsck etc. Use redirection and piping in commands File compression and archiving using tar, gzip, zip, unzip etc. Process commands ps,kill, killall,nice, pgrep, top,htop etc. Network commands ifconfig, ping, netstat, host,ip route etc. Administrator Commands Adduser,Passwd, deluser, usermod, groupadd etc	CO1
2.	Shell Scripting Programming Write a shell script to ask your name, program name and enrollment number and print it on the screen. Write a shell script to find the sum, the average and the product of the four integers entered. write shell script to find average of numbers given at command line Write a shell program to exchange the values of two variables Write a shell program to Print Numbers 1 to 10 using while & do while loop. Write a shell program to Print Numbers 1 to 10 using for loop. Write a shell script to display the digits which are in odd position in a given 5-digit number. Write a shell program to search for a given number from the list of numbers provided using binary search method. Write a shell program to concatenate two strings and find the length of the resultant string	CO1

	Write a shell script to find the smallest of three numbers Write a shell program to count number of words, characters, white spaces and special symbols in a given text	
	Process & Thread Management	
3.	Introduction to C Programming (Statement, Conditional Statement, Loop, Array & Function)	C02
4.	Implement FCFS CPU Scheduling algorithm.	C02
5.	Implement the SJF CPU Scheduling algorithm (For both Pre-emptive and non-pre-emptive).	C02
6.	Implement PRIORITY CPU Scheduling Algorithm (For both Pre-emptive and non-pre-emptive).	C02
7.	Implement Round-Robin CPU Scheduling Algorithm	C02
8.	Implement Multi-Level Queue CPU Scheduling algorithm.	C02
9.	Implement Multilevel Queue CPU Scheduling Algorithm.	C02
	Concurrency and Deadlock Management	
10.	Execute the RACE Condition of Process Synchronization.	C02
11.	Implement the Producer–consumer problem using semaphores.	C02
12.	Design a code and implement the Dining Philosopher problem.	C02
13.	Implement Banker’s algorithm of Deadlock Avoidance.	C02
14.	Execute an algorithm for Deadlock Detection.	C02
	Memory Management	

15.	Implement Contiguous memory variable size partition scheme.	C03
16.	Simulate the First-Fit contiguous memory allocation technique.	C03
17.	Simulate the Best-Fit contiguous memory allocation technique.	C03
18.	Simulate the Worst-Fit contiguous memory allocation technique.	C03
19.	Implement the Non-contiguous Memory Allocation by using Paging.	C03
20.	Implement Contiguous memory variable size partition scheme.	C03
	Page Replacement	
21.	Write a Program to simulate the FIFO page replacement algorithm.	C03
22.	Write a Program to simulate the LRU page replacement Algorithm.	C03
23.	Write a Program to simulate the Optimal page replacement Algorithm.	C03
	Disk Scheduling	C03
24.	Write a program to simulate FCFS Disk Scheduling Algorithm.	C03
25.	Write a Program to simulate the SSTF Disk Scheduling Algorithm.	C03
26.	Write a program to simulate SCAN Disk Scheduling Algorithm.	C03
27.	Write a Program to simulate the C SCAN Disk Scheduling Algorithm.	C03
28.	Write a Program to simulate the LOOK Disk Scheduling Algorithm.	C03
	Modern Operating System	C03

29.	Introduction of CUDA Programming.	C03
30.	Write a program in CUDA print message “Welcome CUDA programming”	C03
31.	Implement matrix multiplication using shared memory in CUDA.	C03
32.	Connects to VMware vCenter and lists all virtual machines along with their power state.	C03
33.	Create a new virtual machine in Azure with specified configurations.	C03
34.	Deploy a simple HTTP-triggered distributed Azure Function.	C03
Total Hours:		48
*Competitive coding list will be shared with the students.		